

MASTER DEMONSTRATION NARRATIVE: “OBSERVE → IDENTIFY → EXPLAIN → CONTEXTUALIZE → EXPLORE → DECIDE”
Target Duration: 3:30–4:00 (Max 5:00)

A disciplined, reproducible live walkthrough of the production platform designed for a multidisciplinary evaluation jury. The goal is **not** to sell future promises or perform ad-hoc exploration, but to guide the jury through a structured spatial screening workflow that proves technical innovation, empirical explainability, and practical decision-support value—anchored strictly by verified scientific guardrails.

MASTER 4-MINUTE PRESENTATION TIMELINE

0:00–0:20 STEP 01 Orient Rondônia Context & ~150k Cells	0:20–1:10 STEP 02 Spatial Risk Field 3D Extrusion & Relative Surface	1:10–1:45 STEP 03 High-Risk Focus Rondônia Arc Hotspot (84%)	1:45–2:30 STEP 04 Explain Risk Signal Dual-Track & TreeSHAP Bars	2:30–3:10 STEP 05 Historical Context 25-Yr PRODES (2001–2025)	3:10–3:40 STEP 06 Scenario Explore Sensitivity Simulation Sliders	3:40–4:00 STEP 07 Decision Close Verification & Value Flow
--	---	---	---	--	--	---

VERIFIED APPLICATION ROUTE & ACTION SEQUENCE

STEP	NARRATIVE PHASE	APPLICATION ROUTE / COMPONENT	PRIMARY PRESENTER ACTION	CORE EVIDENCE DISPLAYED	BUDGET
01	Orient & Scope	index.html (Hero / Status dot)	State study area, problem scale, and platform differentiation	Rondônia boundary; ~150,000 precomputed cells	0:20 (20s)
02	Spatial Risk Field	index.html (Map Controls: 3D/2D)	Rotate 3D Deck.gl view; toggle 2D grid and Hotspot layers	3D risk extrusion (BR-364 corridor); relative risk scale	0:50 (50s)
03	High-Risk Focus	Sidebar Story Tab → Target Btn	Click Target: Highest-Risk Hotspot (auto-fly to Arc Hotspot)	LAT -11.9182, LON -63.8722; 84% Risk; Urgent priority	0:35 (35s)
04	Explain Risk Signal	pages/ai-model.html	Show Dual-Track Architecture & TreeSHAP attribution bars	Track 1 (AUC 0.82) vs Track 2 (AUC 0.7474, 5.0× lift); SHAP	0:45 (45s)
05	Historical Context	pages/historical-intelligence.html	Toggle "Compare with Rondônia" checkbox; click 2004/2012 cards	25-yr official PRODES series (2001–2025); Rondônia vs Basin	0:40 (40s)
06	Scenario Exploration	pages/scenario-simulator.html	Adjust Road slider; toggle Strict Reserve policy switch	Projected risk shift (+2.3% to -4.1%); distribution shift	0:30 (30s)
07	Decision Story Close	index.html (Sidebar Story Steps 04–05)	Highlight targeted verification protocol and 5-node value flow	Targeted inspection planning; structured decision workflow	0:20 (20s)

PRESENTER DISCIPLINE: MUST SHOW VS. SKIP DECISION MATRIX

MUST SHOW (CORE CREDIBILITY)	STEP REF	NICE TO SHOW (IF TIME PERMITS)	SECONDARY	DELIBERATELY SKIP (AVOID RABBIT HOLES)	DO NOT SHOW
3D Spatial Risk Extrusion: Visual prioritization across Rondônia (Steps 01–02). ~150,000 Spatial Scale: Precomputed cell resolution covering the state territory (Step 01). Specific Hotspot Inspection: Rondônia Arc cell (84% score, Urgent priority) (Step 03). Dual-Track Transparency: Track 1 (AUC 0.82) vs Track 2 (AUC 0.7474) (Step 04). TreeSHAP Attribution: Proximity to historical clear-cuts (41.3%) + roads (41.0%) (Step 04). 25-Year PRODES History: Official INPE ground-truth from 2001 to 2025 (Step 05). Decision-Support Purpose: Screening and verification workflow close (Step 07).		Basemap Switcher: Toggle Analytical → Terrain/Satellite basemaps. Risk Trajectory Chart: Point-level 2030–2035 trend canvas in sidebar. Report Export Modal: CSV, GeoJSON, and PDF export trigger buttons. Policy Preset Buttons: "Worst Case" vs "Conservation Focus" buttons in simulator. Jurisdiction Dropdown: Switching from Amazônia Legal to Pará or Mato Grosso. Historical Milestone Cards: Clicking 2004 peak or 2012 historic trough.		Source Code Walkthrough: Do not open IDE or inspect JS libraries unless asked. WebGL Shaders: Avoid deep dives into Deck.gl buffer vertex shaders. Calibrated Probabilities: Never claim a score predicts % chance of future loss. Live Telemetry: Never claim live streaming satellite connections (NASA FIRMS). Live XGBoost Retraining: Sliders are analytical sensitivities, not live fitting. Institutional Deployment: Do not claim formal IBAMA/ICMBio operational integration. Carbon ROI Dollars: Never speculate on dollar values of carbon credits.	

JURY EVALUATION RUBRIC ALIGNMENT

CRITERION 01 Technological Innovation Deck.gl 3D WebGL rendering, 150k precomputed cell architecture, multi-factor GeoAI pipeline.	CRITERION 02 Scientific Credibility Dual-track separation, out-of-time PRODES 2024 testing, TreeSHAP interpretability, strict non-claims.	CRITERION 03 Decision-Support Utility Shifts from passive loss mapping to proactive screening, capacity-constrained ranking, and inspection alerts.	CRITERION 04 Presentation Maturity Tight 4-minute time discipline, verified interactions only, zero unsupported claims, confident Q&A defense.
---	--	--	---

0:00–0:20 (20s)

[index.html](#)

PURPOSE: Establish the project's identity, geographic focus (Rondônia), territorial scale, and fundamental distinction: proactive risk screening vs passive post-loss monitoring.

"This platform organizes environmental spatial data into proactive risk intelligence for Rondônia in the Brazilian Amazon. Rather than mapping deforestation after tree cover is lost, it evaluates 150,000 precomputed cells to surface areas of elevated spatial vulnerability."

SKIP: Code implementation, GitHub repo, or detailed data ingestion mechanics

0:20–1:10 (50s)

[index.html \(btn3d/b...](#)

SHOW: 3D column extrusion where height and color encode relative spatial risk. Highlight high-risk clusters concentrated along Highway BR-364 and expanding agricultural frontiers.

PURPOSE: Demonstrate high-performance WebGL visualization while establishing scientific guardrails: relative risk ranking, not calibrated probability; visual extrusion, not physical elevation.

"The 3D surface visually prioritizes landscape risk. Vertical column height is a visual representation of normalized risk intensity, not physical elevation. Crucially, this represents a relative spatial risk ranking, not a calibrated probability forecast."

SKIP: Speculating on physical tree height or claiming real-time satellite telemetry.

1:10–1:45 (35s)

```
index.html] (focusTo
```

ACTION: Open sidebar Story Tab; click button Target Highest-Risk Hotspot.

SHOW: Smooth camera transition targeting Rondônia Arc Hotspot (LAT -11.9182 | LON -63.8722). Inspect 84% Relative Risk Score, Urgent Priority badge, and driver contribution summary.

PURPOSE: Ground the macro-scale surface into a concrete, auditable point screening event.

"By targeting the highest-priority hotspot, we see exactly why the area receives a higher relative spatial risk score and is surfaced for screening. The platform flags this cell because it sits at the immediate intersection of active clearing frontiers and primary access corridors."

SKIP: Claiming this point is a verified real-world clearing event today without evidence.

1:45–2:30 (45s)

[pages/ai-model.html](#)

ACTION: Navigate to AI Model; scroll through Tree Ensemble to Dual-Track KPI grids.

SHOW: Track 1 Documented Production Benchmark (ROC-AUC 0.82) vs Track 2 Independent Research Model V2 (ROC-AUC 0.7474, 5.0× retrospective enrichment). Show TreeSHAP attribution: Historical Loss (41.3%) and Road Proximity (41.0%).

PURPOSE: Prove machine learning interpretability and transparent scientific integrity.

"The system does not treat AI as a black box; it exposes the spatial signals behind every score. We strictly separate the documented 0.82 production benchmark from our independent research evaluation, where TreeSHAP shows that historical clear-cuts and road corridors drive over 80% of model attribution."

SKIP: Math derivations of tree boosting; do not claim Model V2 is running live in production.

2:30–3:10 (40s)

[pages/historical-in...](#)

ACTION: Check Compare with Rondônia ('hiCompareRondonia'); click 2004/2012 callouts.

SHOW: 25-year official PRODES time series (2001–2025). Compare Amazônia Legal (5,731 km² in 2025) with Rondônia benchmark (647 km²). Highlight historical inflection points.

PURPOSE: Contextualize risk predictions within sovereign historical environmental monitoring.

"Spatial risk cannot be evaluated in isolation. We place the risk layer inside a 25-year historical monitoring continuum from INPE PRODES, allowing analysts to compare current spatial vulnerability against multi-decade basin trends and policy shifts."

SKIP: Lengthy political commentary on past Brazilian presidential administrations.

3:10–3:40 (30s)

[pages/scenario-simu](#)

ACTION: Move Road Expansion slider (`roadSlider`); toggle Strict Forest Reserve.

SHOW: Dynamic shift in Projected Regional Risk (+2.3% to -4.1%) and risk distribution chart.

PURPOSE: Demonstrate exploratory policy sensitivity simulation with strict scientific guardrails against false forecasting claims.

"The simulator allows decision-makers to explore policy assumptions through analytical sensitivity modeling. We explicitly present this as exploratory sensitivity simulation, not live browser-side XGBoost retraining or deterministic forecasting."

SKIP: Claiming this predicts exact hectares or financial carbon offset credit revenues.

3:40–4:00 (20s)

[index.html \(Sidebar...](#)

ACTION: Return to index.html; highlight Decision Story Step 04 & 05 Value Flow.

SHOW: Inspection planning window and decision-support value flow toward verification planning.

PURPOSE: Conclude with a memorable, grounded statement defining the platform's true utility.

"The purpose is not to replace established forest-monitoring systems. It is to organize spatial evidence into an interpretable workflow for risk screening, prioritization, contextual analysis, and verification planning."

SKIP: Overpromising guaranteed real-world patrol enforcement cost reductions

DEMONSTRATION FAILURE SAFETY & 4-TIER FALLBACK PROTOCOL

FAILURE SCENARIO	TRIGGER / SYMPTOM	IMMEDIATE TECHNICAL PIVOT	PRESENTER SCRIPT & BEHAVIORAL PROTOCOL
Tier 1: WebGL Crash / GPU Context Lost	3D map canvas turns black or Deck.gl fails to initialize.	Click 2D button ('btn2d') or Analytical basemap ('styleDark'). 2D canvas renders reliably without GPU instancing.	"Let us switch instantly to our high-contrast 2D spatial screening view. This isolates the spatial risk field across the 150,000 cells without requiring 3D WebGL depth buffers." (Keep momentum, no apology).
Tier 2: Network Throttling / Slow Basemap	Mapbox satellite tiles load sluggishly or show blank grey grid.	Toggle to Analytical style ('styleDark'). The vector polygon grid is precomputed and cached locally in memory.	"The precomputed 150,000 risk cells are fully cached in browser memory, enabling analytical spatial screening independent of external satellite tile delivery."
Tier 3: Full Browser Freeze / Tab Crash	Browser becomes unresponsive during 3D camera transitions.	Pivot immediately to pre-opened fallback tab or high-resolution PDF dossier (GeoAI_Technical_Evidence.pdf).	"As documented in our technical evidence dossier, the identical precomputed risk field can be examined directly here in our published submission records." (Never claim a static PDF is live data).
Tier 4: Interaction / Slider Stutter	Simulator sliders or animation loops hitch or lag.	Skip simulator; pivot directly to Step 02 & 03 of the Decision Story sidebar in index.html.	"The analytical sensitivity model demonstrates that road proximity drives +41% of risk expansion, as verified in our sidebar feature attribution panel."

PRESENTER EXECUTION RULES FOR LIVE BROWSER DEMONSTRATIONS

Rule 01: Pre-Load Exact Tabs Before entering the jury room, pre-load three browser tabs: (1) index.html, (2) pages/ai-model.html, and (3) pages/historical-intelligence.html. Avoid live reloads.	Rule 02: Never Bluff Glitches If an interaction fails, acknowledge it with professional poise and pivot to the next verified view. Never pretend a static fallback is streaming live data.	Rule 03: Anchor on Locked Metrics When challenged on numbers, cite locked submission metrics: ROC-AUC 0.82 (Track 1) and ROC-AUC 0.7474 with 5.0× retrospective enrichment (Track 2).
---	--	---

JURY DEFENSE: TOP 5 TECHNICAL CHALLENGE QUESTIONS

Q1: "Is this model predicting the exact probability of deforestation occurring next month?" Defense: "No. We explicitly do not claim calibrated probability forecasts. Because deforestation is an extreme rare event (0.14% annual prevalence in the test lattice), Raw scores are interpreted as relative spatial risk rankings across the precomputed cells, not as calibrated probabilities."
Q2: "What does the 5.0× lift metric actually mean, and how was it measured?" Defense: "The 5.0× metric represents retrospective top-decile enrichment evaluated out-of-time on official PRODES 2024 clear-cuts. When inspecting the top 10% highest-ranked test cells, the model captured 5 out of the 10 observed clearing events (50% recall), which is exactly 5.0 times higher than random inspection (10%). It is not an operational patrol field measurement."
Q3: "Does your platform ingest live real-time satellite telemetry or NASA FIRMS alerts?" Defense: "No. The current production application operates on 150,000 precomputed spatial cells derived from authoritative annual INPE PRODES, IBGE, and infrastructure records. Real-time satellite alert integration is documented as a future deployment phase, not an active prototype feature."
Q4: "Why does historical loss proximity dominate the model over road infrastructure in Model V2?" Defense: "In this evaluation sample, proximity to historical loss was the strongest single-feature benchmark, suggesting strong spatial persistence in the observed clearing pattern. This is an empirical finding, not a universal causal claim."
Q5: "Why did you develop Model V2 instead of simply reproducing the original 0.82 benchmark?" Defense: "The original 0.82 benchmark is documented from the project's inception, but its training labels were unrecorded in the repository. Rather than manufacturing synthetic pseudo-labels to fake a reproduction, we exercised scientific discipline : locking the 0.82 benchmark as Track 1 and conducting an independent, reproducible out-of-time evaluation (Model V2) as Track 2."
CONFIDENT TRANSITION INTO JURY Q&A <i>"Thank you. We have structured this platform to prove that AI in conservation is most valuable when it provides transparent, interpretable decision support within rigorously defined scientific boundaries. I welcome your questions."</i>